



Lab 10 Tasks Report Submission

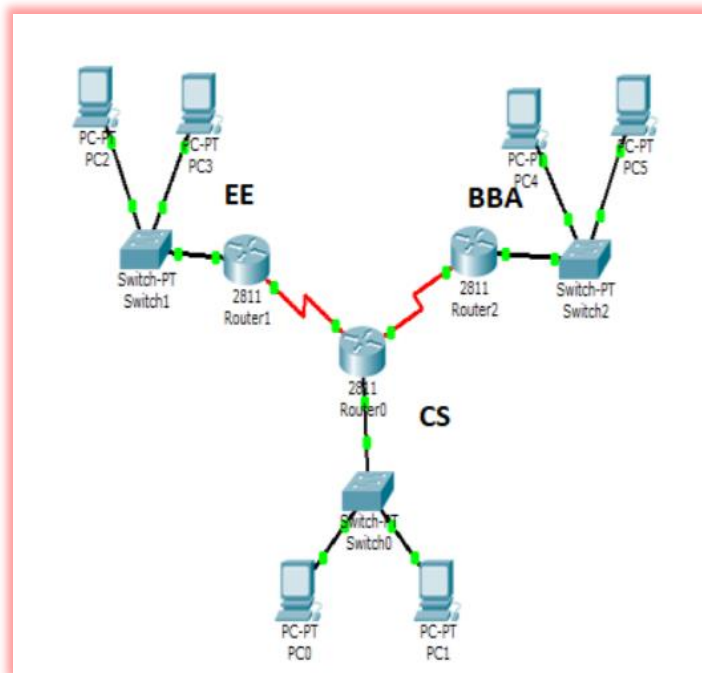
Task-01

1. Implement the above given example on cisco packet tracer but use network address as 10.20.0.0/24 using RIP.
2. Let's consider an example of subnetting for ABC Company. There are 3 departments i.e. Finance, HR and Development.
You have to perform subnetting for the allocation of the given requirement:
120 PCs for Development
35 PCs for Finance
10 PCs for HR

The network address for the given scenario is 200.16.100.0/24. Only show calculation.

3. Let's consider an example of subnetting for FAST NUCES. There are 3 departments i.e. CS, EE and BBA.
You have to perform subnetting for the allocation of the given requirement:
90 PCs for CS
50 PCs for EE
20 PCs for BBA

The network address for the given scenario is 195.168.10.0/24. Implement it on Cisco Packet Tracer and Apply RIPv2.





Task-02

Nadeem, a network engineer at a small IT firm, has been tasked with helping his colleagues Sameer and Abdullah set up a dynamic routing network using RIP version 2 in Cisco Packet Tracer. The goal is to enable seamless communication between different office departments spread across three routers. Each team member is responsible for configuring one router in the topology.

The network consists of three routers (RouterNadeem, RouterSameer, and RouterAbdullah), each connected to separate LAN segments representing HR, Finance, and IT departments. The LAN segments use the IP ranges 192.168.10.0/24, 192.168.20.0/24, and 192.168.30.0/24, respectively. The routers are connected to each other through serial links using IPs in the 10.0.x.0/30 range. All routers must advertise their directly connected networks using RIPv2, ensure route updates are being exchanged, and verify network convergence.

Tasks:

1. Basic Configuration:

Assign IP addresses to all devices and interfaces according to the topology. Configure hostnames on each router as RouterNadeem, RouterSameer, and RouterAbdullah. Ensure that the LAN PCs can successfully ping their default gateways.

2. Implement RIPv2 Routing:

Enable RIPv2 on all three routers. Use appropriate network commands to advertise both LAN and WAN networks. Ensure that version 2 is explicitly set and no auto-summary is used to prevent classful summarization.

3. Connectivity Testing:

After RIP is configured, verify that routing tables on each router include all LAN networks. Test end-to-end connectivity by pinging from a PC in Nadeem's HR department to a PC in Abdullah's IT department.

4. Troubleshooting & Optimization:

Sameer notices that his router is not receiving updates from Abdullah's router. Investigate the issue and document the solution. Additionally, configure RIP passive interfaces on LAN ports to enhance security by preventing unnecessary RIP advertisements on internal segments.

Task-03

Implement the network topology described below in Cisco Packet Tracer and configure dynamic routing using the RIP protocol. Ensure proper IP addressing, router configuration, and successful inter-network communication through RIP-based route advertisement.

